Review

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Old Strategies, New Environments: Reinforcement Learning on Social Media

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ABSTRACT

The rise of social media has profoundly altered the social world, introducing new behaviors that can satisfy our social needs. However, it is not yet known whether human social strategies, which are well adapted to the offline world we developed in, operate as effectively within this new social environment. Here, we describe how the computational framework of reinforcement learning (RL) can help us to precisely frame this problem and diagnose where behavior-environment mismatches emerge. The RL framework describes a process by which an agent can learn to maximize their long-term reward. RL, which has proven to be successful in characterizing human social behavior, consists of 3 stages: updating expected reward, valuating expected reward by integrating subjective costs such as effort, and selecting an action. Specific social media affordances, such as the quantifiability of social feedback, may interact with the RL process at each of these stages. In some cases, affordances can exploit RL biases that are beneficial offline by violating the environmental conditions under which such biases are optimal, such as when algorithmic personalization of content interacts with confirmation bias. Characterizing the impact of specific aspects of social media through this lens can improve our understanding of how digital environments shape human behavior. Ultimately, this formal framework could help address pressing open questions about social media use, including its changing role across human development and its impact on outcomes such as mental health.

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In recent years, social media have dramatically reshaped our daily lives: almost half of U.S. teenagers now report being online almost constantly (1). Such changes have introduced novel ways to meet human social needs. We can interact with huge numbers of friends at almost any time; communicate in new ways such as "liking" messages or brazenly ignoring them; and gather social information, such as what our friends are doing, in unprecedented detail. However, human social behaviors developed over long evolutionary timescales and adapted to offline environments. The utility of our social strategies in the increasingly digital world, which is governed by radically different rules and designs, is poorly understood.

Amid mounting concerns about the role of social media in youth mental health declines, improving this understanding is imperative (2–4). In this review, we argue that the framework of reinforcement learning (RL), which is imported from cognitive and computer science, is a powerful tool with which to do so (5–7). RL characterizes the social world as a reward-maximizing problem: we must learn the best way, in a given environment, to obtain the social rewards that we need (6). This perspective helps uncover the characteristics of offline environments to which human social strategies adapted, thereby pinpointing the relevant aspects of the social world that online environments have changed. Building on decades of RL research, we can then better understand impacts of social media through changes to how we pursue and consume social rewards in the digital age (8).

SOCIAL REWARDS AND SOCIAL MEDIA

Humans have an innate need for affiliation (9). Social connection provides adaptive benefits ranging from safety to mating, while weak social ties predict poor mental and physical health (10,11). We can identify and regulate social needs using evolutionarily conserved systems, a process known as social homeostasis (12). But the drive for sociality is complex and abstract. To motivate daily social behaviors, we assign value to proximal targets: learned behaviors and stimuli that maintain social connection, such as smiling or helping others (13). While these social rewards provide the ingredients for achieving affiliation, they also make us feel good in and of themselves (Box 1) (14).

The ways we seek and receive social rewards today differ markedly from the distant and even recent past. One of the most rapid and far-reaching changes is digitalization (15–18). The online social world differs from the offline one in important ways, which we can conceptualize in terms of the digital affordances of online environments (Table 1). These are the abstract action possibilities that features of social media platforms allow—for example, quantification of social feedback via likes or views (19–22). Digital affordances enable unique forms of social interaction, while rewarding behaviors in offline environments often lack exact online analogs (21). As the social world moves progressively online, it is essential to understand how these changes interact with reward learning.

A COMPUTATIONAL PERSPECTIVE ON SOCIAL REWARD LEARNING

Imagine that you want to gain the social reward of popularity. How can you achieve this? The insight that such human reward pursuit can be studied with objective scientific methods

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Box 1. Glossary

Action probabilities: The probability of performing action *a* is the likelihood that the RL agent will perform action *a* at the action selection stage. The summed probabilities of all possible actions are equal to 1.
Action selection: The third and final stage of the RL process, where probabilities of performing each potential course of action are computed.
Confirmation bias: A learning bias wherein a higher weight is given to new information that is consistent with our existing beliefs when updating our expectations.
Context-dependent learning: The tendency to scale updated reward expectation relative to the context rather than updating reward expectation consistently across contexts according to absolute value.
Computational phenotype: An individual's typical cognitive processes characterized using a set of parameters obtained from fitting a computational model to their behavioral data.
Delay discount: The tendency to value immediate or sooner rewards more highly than rewards farther in the future.
Description-experience gap: The tendency for learning to be shaped differently by equivalent information depending on whether it is directly experienced or explicitly described.
Digital affordances: The perceived action possibilities of digital environments, which are jointly determined by features of digital platforms and users' perceptions of those features.
Directed exploration: Taking actions that are specifically intended to reduce uncertainty, a form of exploration that can be distinguished from random exploration.
Effort cost: The subjective cost of physical or mental effort associated with a particular action.
Expected reward: The expected reward for action *a* is the long-term expected reward that can be gained from action *a*, including both immediate and expected future outcomes. Specifically, the expected reward for an action is equal to the size of the reward multiplied by the likelihood of receiving the reward from the action.
Exploration-exploitation dilemma: The trade-off between exploiting what we already know by choosing the highest valued option versus exploring by trying out new options that may lead to more rewarding outcomes.
Goal-directed system: A behavioral system that underlies purposeful actions in pursuit of rewards by altering the policy in response to experienced rewards or punishments.
Habitual system: A behavioral system that underlies the automatic repetition of actions (habits) even if they are no longer rewarding.
Matching law: Hernstein's (1961) matching law provides a robust, descriptive account of how animals, including humans, perform actions at a ratio that matches the relative ratio of rewards available for each action.
Model-based RL: An RL system that updates value based on internal models of the environment and then selects actions based on expected future outcomes. Adapts quickly to changing environmental contingencies but requires more computational power or effort than model-free RL.
Model-free RL: An RL system that updates value directly based on action outcomes. Adapts slowly to changing environmental contingencies but requires less computational power or effort than model-based RL.
Negativity bias: A form of valence asymmetry where negative information is weighed more heavily than positive information.
Proximal rewards: Rewards that acquire learned rewarding value by association with ultimate goals.
Random exploration: Acting randomly to learn from the consequences of new actions, a form of exploration that can be distinguished from directed exploration.
Risk aversion: The tendency to prefer options with more certainty over riskier options, even when the expected reward is equal.
Selective attention: A preferential focus on certain environmental information, intended to be on information that is deemed to be more goal relevant.
Social homeostasis: The ability of humans to monitor the quantity and quality of their social contact, compare it with a set point that defines their needs, and perform actions to match their social contact to the set point. A theoretical extension of the concept of physiological homeostasis.
Social rewards: Stimuli to which we learn to assign value because they serve the long-term goal of social connection and affiliation but which also feel good in and of themselves.
Subjective value: The utility assigned to a particular action after integrating expected reward with subjective discount factors such as effort cost and delay discount.
Ultimate goals: Rewards that are inherently desirable and rewarding, such as intimacy.
Update: The first stage of the RL process, during which new information from the environment is combined with prior expectations to update the expected rewarding outcome of each possible action.
Valence asymmetries: Differential weighing of information that is better versus worse than what we expected when updating our expectations. Valence asymmetry can occur in both the positive direction (positivity bias) and negative direction (negativity bias).
Valuation: The second stage of the RL process, when the expected rewarding outcome of the action is subjectively discounted by the subjective cost of effort and temporal delay.

RL, reinforcement learning.

can be traced to behaviorism—a leading psychology movement in the 1920s (23). A fundamental tenet of behaviorism is that behavior is determined by positive and negative environmental reinforcement (24). Leading behaviorists, such as Skinner and Bandura, inspired a profusion of research into how reinforcement schedules and environmental stimuli shape actions (24,25). For example, Hernstein's matching law describes how animal response rates match the relative rate at which each response is rewarded (or reinforced) (26).

Such descriptive accounts have been influential in quantifying relationships between environmental stimuli and behavior, but they do not explain why such relationships occur (27). In contrast, RL synthesizes descriptive findings into a normative framework. This characterizes diverse reward-seeking strategies as appropriate, interacting parts of a systematic process with a common goal: maximizing reward.

We can conceptualize the RL process within a 3-stage value-based decision-making framework (28) (Figure 1). In

Table 1. Taxonomy of Social Media Affordances Most Relevant to the RL Process

Social Media Affordance	Definition	Example Feature	Example Interaction With RL Process	Related Approaches in Previous Literature
Asynchronicity	The degree to which an interaction is experienced as occurring in real time	Messages and posts may be viewed and responded to much later than when originally posted.	Increases social uncertainty—for example, over whether a friend's lack of reply results from anger or from not having seen the message (36), which can alter the balance between model-based and model-free RL.	Asynchronicity (15,36), synchronicity (19,94), although distinct from the concept of media synchronicity used in (135)
Availability	How easily a user can access the technology; how easily the user can be reached through the technology	Fast-forward option for video viewing, social media applications for mobile phones	Decreases the time required to gain a social reward, increasing the value of quickly obtainable online rewards due to delay discount	Availability (19,21,136), accessibility (137,138), connectedness (139)
Bandwidth	The breadth of socioemotional cues transmittable by messages	Message reactions, emojis, gifs, text limits, voice notes, and read receipts reduce bandwidth	Low-bandwidth messages are difficult to disambiguate, which could increase the uncertainty of the expected reward, increasing directed exploration.	Bandwidth (19,140)
Editability	The degree to which messages and posts can be crafted, refined, and edited	Beauty filters, video editing tools	Alters the range of content that we view, changing our subjective value of attributes via context-dependent learning (135)	Editability (19,141), rehearsability (135)
Persistence	The degree to which messages and posts remain accessible to senders and/or receivers in their original form	Settings that automatically delete messages enable ephemeral interactions with low persistence, while searchable chat history without deletion enables high persistence.	Increases the risk associated with mundane social actions, interacting with valence asymmetry	Persistence (19,22,141), permanence (21,137), retrievability (142), reprocessability (135)
Personalization	The degree to which information is tailored (e.g., by recommendation algorithms) to fit the identity, preferences, or expectations of the receiver	Feed algorithms, blocking followers	Redistributes the information that we consume to disproportionately confirm our beliefs, interacting with confirmation bias	Personalization (19,140,143), curation (144)
Quantifiability	How countable social information is	Likes, views, follower counts	Quantifiable rewards require low effort to transmit (e.g., a simple button press), reducing the effort required for social interaction, interacting with the RL effort cost.	Quantifiability (19,21), paralinguistic digital affordances (145)
Replicability	The ability to duplicate messages and forward them on to others	Features that enable forwarding and preserving of messages, such as via screenshots or forwarding features	Increases the risk associated with mundane social actions, interacting with valence asymmetry	Replicability (19,22)
Scrollability	The ability to engage in continuous scrolling, also referred to as the infinite scroll	Infinite scroll newsfeeds	Induces a stable context that increases the weight given to the habitual system (72)	Scrollability (75), infinite scroll (72)
Triggered Attending	Responding to online content when an alert tells the user to do so	Audible phone notifications	Directs selective attention toward the most salient stimuli, even if these are not the most goal relevant	Triggered attending (20,146)
Variable Reward Schedules	The degree to which social interaction and feedback occur on variable time schedules	Phone notifications at arbitrary times of the day, newsfeeds with unpredictably rewarding posts	Increase the weight given to the habitual system (24)	Variability of social rewards (19), reinforcement schedules (113)

Table 1. Continued

Social Media Affordance	Definition	Example Feature	Example Interaction With RL Process	Related Approaches in Previous Literature
Visibility	The relative ease with which messages can be located and seen by others	Public posts, screenshots	Increases the risk associated with mundane social actions—for example, a mundane interpersonal gesture such as sending a joke to a friend is accompanied by increased risk of public humiliation. Interacts with valence asymmetry	Visibility (19,141), scalability (15,22), publicness (21)

Social media affordances discussed throughout this review are included, along with examples of features through which each affordance is enabled, ways in which each affordance interacts with the RL process, and previous research discussing each affordance.

RL, reinforcement learning.

the case of attaining popularity, for example, stage 1 is to use any new rewards (or punishments) that you have gained from acting in your social environment to update your expected reward from each possible action. Expected reward is the size of the anticipated reward multiplied by the likelihood of receiving it. For example, after noticing that giving thoughtful gifts raises your popularity, you increase your expected reward from this action.

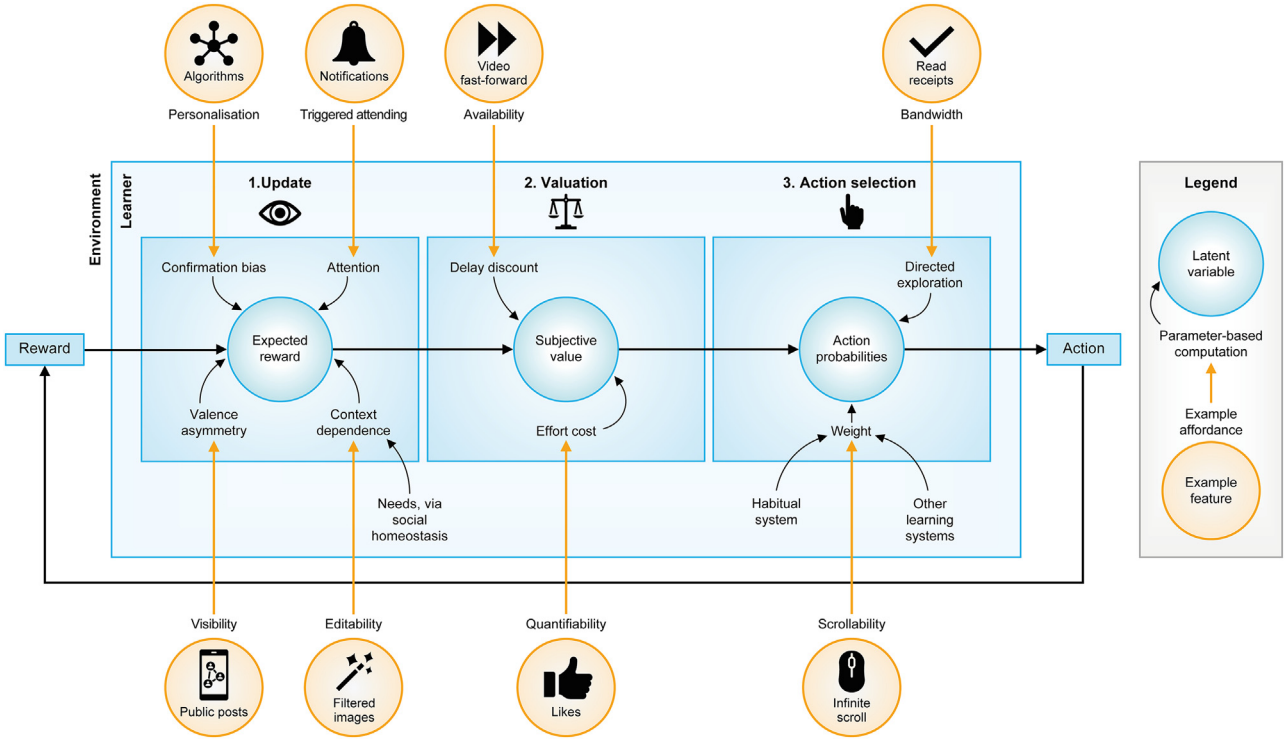

Figure 1. Interaction between social media affordances and the reinforcement learning process. This diagram maps out the reinforcement learning process, showing how affordances and associated features of social media interact with parameter-based computations at each of the 3 stages. To simplify the figure for visualization, one example affordance and one example feature are shown for each parameter-based computation, although in reality, multiple affordances and features are relevant to each step, and each affordance can interact with multiple stages of the reinforcement learning process. At each of the 3 stages, a latent variable is constructed that is passed into the calculation at the next stage. At the update stage, the latent variable expected reward is constructed by combining reward expectation at the previous step with newly experienced reward. This update is modulated by attention, biases, and context-dependence, each of which are sensitive to changes in social media environments. Context is both internal (e.g., an individual's needs to maintain social homeostasis, such as a heightened need for social connection in adolescence) and external (e.g., social media content viewed, such as body-positive images). At the valuation stage, a subjective value is assigned to each potential course of action by applying delay discount and effort costs to reduce the perceived value of rewards that are farther in the future or higher effort. Finally, at action selection, action probabilities are assigned to each action by integrating the values with other priorities such as exploration and relying on habits. When the subsequent action is performed, this interaction with the environment creates a new rewarding (or punishing) outcome, which is then combined again with the previous expected reward, and the process continues.

Box 2. The RL Process

The RL process can be thought of as encompassing 3 stages. At each stage, a new variable is constructed using parameter-based computation, which is then passed into the equation at the next stage.

1. **The update stage**

$$Q_a \leftarrow Q_a + \alpha(r(R) - Q_a) \tag{1}$$

At the update stage, the expected long-term reward for action a (specifically, the size of the reward multiplied the likelihood of receiving it), Q_a, is updated by combining the previous expected reward with newly experienced reward R. R is first scaled according to the context-dependence function, $r(\,)$, which integrates both external context (e.g., range of stimuli viewed) and internal context (e.g., heightened need for social rewards due to developmental needs in adolescence). Then, Q_a is updated proportionally to the prediction error, $r(R) - Q_a$, which is the difference between the expected and experienced reward. The learning rate, α, parameterizes the weight that is given to the new information relative to the previous expectation in this update. The update is distorted, exhibiting valence asymmetry and confirmation bias. Each of these can be instantiated in various ways algorithmically; for example, a valence asymmetry can be instantiated via separate α for positive versus negative prediction errors.

2. **The valuation stage**

$$d(Q_a) \leftarrow \frac{Q_a}{1 + (\kappa_E * E_a + \kappa_D * D_a)} \tag{2}$$

At the valuation stage, the discounted subjective value for action a, $d(Q_a)$, is calculated by integrating the newly updated expected reward for action a, Q_a, with costs such as the effort for action a, E_a, and the delay of the reward, D_a. The parameters κ_E and κ_D parameterize the size of the effort cost and delay discounting, respectively, which vary across individuals.

3. **The action selection stage**

$$P_a \leftarrow s[(1 - \omega) * d(Q_a) + \omega * H] \tag{3}$$

Finally, $d(Q_a)$ is integrated with other long-term priorities to allocate a probability, P_a, of performing action a. The weight, ω, determines the relative reliance on different behavioral systems (here, the habitual vs. goal-directed system), a weight that varies both across individuals and contexts and is modulated by environmental affordances. H is the action recommended by the habit system, although this framework can be extended to encompass different systems. The function $s[\,]$ governs the probability of selecting the action versus exploration, which may be random or directed exploration.

RL, reinforcement learning.

Next, you use this updated expectation in stage 2, valuation. Here, you assign a subjective value to each option, weighing expected rewards with costs. For example, the value of giving thoughtful gifts is reduced because it requires substantial effort and time compared with alternatives such as making a phone call [due to the effort cost and delay discounting of future rewards (27,29)].

Finally, in stage 3, you decide how to act. During action selection, you determine action probabilities, i.e., your probability of actually performing each action. Perhaps the action with the highest subjective value, even after accounting for effort and time, is to give thoughtful gifts. You may choose this highly valued action, but you may also perform novel actions to learn more about the world, or habitual actions, repeating what previously worked without recalculating value each time. In the end, you may decide to do something completely new, such as organizing a dinner party, to see whether this provides even greater reward than giving gifts or phone calls.

In this way, RL unifies social behavior into a teleological process. Specifically, social RL can be understood as a motivational component of social homeostasis, with the reward defined using how close an action gets you to a homeostatic set point (your social need) (12,30). In what follows, we explore how online environments interact with, and at times exploit, features of RL at each stage: update, valuation, and action selection. Subsequently, we discuss how this approach can enhance the study of social media use.

THE UPDATE STAGE

At the first stage, RL agents—for example, humans who are pursuing social rewards—learn from new experiences to update their expected reward for each possible action (Figure 1 and Box 2).

Proximal Rewards

First, RL agents must establish what to conceptualize as rewarding. Ultimate goals such as affiliation and intimacy are inherently rewarding. But people also learn to value proximal rewards, which are stimuli that acquire value by association with ultimate goals (13,31,32). For example, praise is a proximal reward because it reliably predicts the ultimate goal of social connection.

Because social needs often require complex chains of events before satisfaction, proximal rewards are a potent mechanism to motivate long-term goal accomplishment (33). However, this is most adaptive if proximal rewards are consistent predictors of ultimate goals. Social media have introduced a plethora of new proximal rewards such as likes, followers, and views: merely viewing likes on others' photos activates neural reward circuitry in adolescents (34). Qualitative research has shown that although adolescents often ascribe emotional meaning to likes they receive, these likes sometimes reflect minimal effort or interest from their senders (35). Thus, digital affordances such as high availability and low bandwidth may decrease the certainty of the associative relationship between proximal rewards and ultimate goals (36) (Table 1). This could render online reward-seeking behaviors less consistently quenched by the stimuli that originally reinforced them (e.g., genuine intimacy).

Selective Attention

When encountering a reward, humans use selective attention to focus on the most relevant new information (37,38).

Selective attention is beneficial if attention-attracting phenomena are also the most important for reward attainment. However, social media platforms manipulate attention, with clickbait content and audible phone notifications drawing users online at arbitrary times (triggered attending, Table 1).

Manufactured notification structures can artificially decouple salience from goal relevance, interrupting offline socializing—for example, a family dinner—with distractors from the online world. Because phone use during offline socializing can lower feelings of social connection in both adults (39) and adolescents (40), this may reduce the quality of offline social experiences. But notifications can also help offload monitoring of the social world to external stimuli; disabling phone notifications can increase fear of missing out (41) and phone checking frequency (42) in adults.

Ultimately, whether attention-attracting online stimuli have positive or negative effects may depend on their structure. Batching notifications three times a day improves adult psychological wellbeing compared with notifications both at random times and not at all (43). Overall, aligning triggered attending with our priorities could maximize the benefits, while minimizing the pitfalls, of selective attention in online environments.

Context-Dependent Learning

Humans value new rewards on a relative rather than an absolute scale, a process known as context-dependent learning (44–46). Specifically, when evaluating features such as attractiveness and body shape, we adapt our appraisal to the range of stimuli that we have encountered. On social media, the range of people to whom we are exposed extends beyond our social circle and includes altered attributes via filters and curated profiles (editability, Table 1). Viewing idealized body images on social media can negatively affect young adults' own body satisfaction (47,48).

Such impacts have caused much concern, and different mechanisms of counteracting them have been explored. For example, viewing body-positive images increases young women's body satisfaction (47). However, flagging idealized images with explicit disclaimers describing them as unrepresentative does not have these mitigating effects (49). At first, it may seem to be a logical discrepancy that two ways of delivering equivalent information have different effects. However, RL resolves this by attributing such effects on body satisfaction to context-dependent learning, a low-level feature of human cognition that is shaped more by experience (e.g., viewing body-positive images) than description (e.g., disclaimers). This is known as the description-experience gap (46) (Box 1).

Learning Biases

When updating expected reward, humans do not weigh all new information equally. Instead, the update is systematically biased: For example, we weigh positive and negative information differently (valence asymmetry) and weigh information that confirms what we already believe more highly (confirmation bias) (50). Paradoxically, these biases are beneficial for learning across a range of conditions (50,51). However, social media environments may violate the statistical conditions that make these biases advantageous.

One benefit of valence asymmetry is that it enables RL agents to treat risky choices differently from certain ones[1] (52). Valence asymmetry can favor either positive or negative outcomes in different contexts (53), but people typically exhibit risk aversion, i.e., preferring safe over risky options (54,55). Digital affordances such as replicability, visibility, and persistence heighten the risk associated with even ordinary interactions, such as a flirtatious remark, increasing the risk of virality and humiliation (Table 1). Combined with risk aversion, such affordances may therefore discourage previously innocuous social actions online. Importantly, the ability to optimally adjust valence asymmetry to levels of risk improves across development (53,56), suggesting that younger people may have particular difficulty adapting to changed risk on social media.

Like valence asymmetry, confirmation bias is adaptive in many contexts (57). However, social media algorithms can specifically construct the information that we receive to reinforce existing beliefs (personalization, Table 1) (58,59). One study simulated RL agents with confirmation bias to investigate how algorithmic content streams interact with biased RL (60). The researchers found that disproportionately feeding confirmatory information to the simulated agents (i.e., emulating a personalized algorithm) amplified their attitude polarization. Thus, RL simulations can trace the roots of even societal-level social media impacts, such as political polarization, to evolved reward-learning strategies. This method could inform efforts to better align algorithms with our psychological predispositions (59).

THE VALUATION STAGE

After updating, RL agents integrate expected reward with other constraints and priorities to assign subjective values to courses of action (Figure 1 and Box 2).

Discounting

Both the time and the effort required to obtain a reward decrease subjective value. These propensities, delay discounting and effort cost, help maximize reward per unit time while conserving energy (27,29). Social media, through affordances of quantifiability and availability (Table 1), enable users to give and receive vast numbers of social rewards at minimal effort and at pace. For example, simply opening social media on your phone can reveal likes from multiple friends.

For some, this increased availability of social rewards may interact with deficits in offline social reward to increase overall gain (17). For example, a qualitative study found that rural LGBTQ (lesbian, gay, bisexual, transgender, queer) youths used social media to connect with like-minded communities, thereby compensating for the paucity of offline social rewards in their isolated surroundings (61).

[1]This is one of the various, non-mutually exclusive ways that the RL process can account for risk: For example, at the valuation stage, risk may also affect the evaluation of outcomes via a nonlinear utility function across gains and losses (54).

However, favoring quick, low-effort rewards may be maladaptive if the reward distribution is highly skewed, such that one type of reward (e.g., online reward) is significantly quicker and easier to obtain. Some social media companies take formidable measures to increase online reward rate: for example, TikTok's 2× playback button allows users to fast forward videos at double speed (62). Therefore, it can be hypothesized that the most available proximal rewards in the next few seconds are often on social media, even when these are less conducive to the fulfillment of ultimate goals. Consistent with the possibility that discounting displaces offline with online time, time spent on smartphones correlates with delay discounting in adolescents (63).

THE ACTION SELECTION STAGE

The final RL step involves action selection, translating subjective values into a decision about how to act (Figure 1 and Box 2).

Different Behavioral Systems

When selecting an action, humans integrate information from multiple different behavioral systems that dictate behavior (64–66). Classical psychology makes a key distinction between the goal-directed system and the habitual system (67,68). Goal-directed behavior is reflective and motivated, with actions being adjusted in response to experienced rewards to pursue goals. For example, children switch strategies fast after an unexpected response from a partner in a cooperative game (69). In contrast, the habitual system automatically repeats actions that were rewarding in the past, which is less flexible but can save mental energy (68). For example, adults habitually choose social partners who previously offered high rewards, even when the partner no longer offers reward (70).

The trade-off between different behavioral systems helps to efficiently maximize reward as environmental demands change. But certain digital affordances disproportionately increase the weight given to the habitual system (67,71–74). Social media notifications arrive on variable reward schedules, which are known to effectively maintain vigorous habitual behavior (Table 1) (24). Furthermore, social media user interfaces induce stable contexts via affordances such as scrollability (e.g., the infinite scroll), which also encourage habits (Table 1) (72,75). Some social media interventions aim to disrupt these affordances (76,77), for example intervening when an application is opened to decrease context stability and therefore reduce the weight given to the habitual system (78).

RL offers more precise and granular distinctions between different behavioral systems than the goal-directed versus habitual dichotomy (79,80). This is useful because different RL systems offer divergent predictions about how best to disrupt unhealthy behavioral patterns (83). Model-based RL updates expected reward using an internal model, while model-free RL updates expected reward based on which actions are typically rewarding (79,81). Both RL systems are distinct from a value-free process that does not update reward but simply repeats actions when in a familiar context (82).

For example, take the case of deciding when to post on social media. Model-based RL might try to maximize rewarding likes by internally modeling friends' minds and posting when they are most likely to like it. In contrast, model-free RL might plan in less depth, simply using trial and error to post more frequently when more likes have been received recently. This has been modeled by adapting animal models of response speed to the social media domain (6,83–85). Finally, value-free processes may favor posting whenever in a particular context regardless of expected reward (82,86).

A higher weight is given to model-free RL when the predictions of model-based RL are more uncertain and vice versa (87,88). Therefore, social media could alter the balance between model-free and model-based RL. For example, via low-bandwidth features such as likes, social media increases uncertainty of internal models of what our friends are thinking (Table 1) (36). This could decrease the weight given to model-based RL, potentially explaining recent findings that posting latency on social media is well characterized by model-free RL (6,83–85). Different RL systems that underlie posting suggest different ways to change behavior: For example, explaining that someone's friends are currently resentful and therefore less likely to like a post might affect posting frequency based on model-based but not model-free RL.

However, value-free processes can also govern social media posting: For example, more frequent Facebook posters reduced their posting rates following a 2007 platform design change and were also less likely to increase posting rate after receiving likes (86). So, reminiscent of value-free processes, their posting was more influenced by the context (Facebook user interface) than a model-free RL process responding to numbers of likes. These frequent users also self-reported being more habitual (86), which supports theoretical work mapping psychology's goal-directed system onto both model-based and model-free RL and the habitual system onto value-free processes (82) (Box 3). However, how computational systems map onto the classical goal-directed versus habitual distinction remains the subject of active research (79).

Exploration

As well as integrating diverse systems, RL agents must decide between taking a highly valued action or exploring to gather new information—the exploration-exploitation dilemma. Humans engage in either random or directed exploration, respectively adding randomness to choice behavior or actively directing choice toward more uncertain options (89–91), with exploration strategies changing from childhood to adulthood (91–93). Online environments modulate social uncertainty via affordances such as low bandwidth and asynchronicity (Table 1) (36,94). Specifically, many observations on social media, such as that a friend has "viewed" your content, provide ambiguous insight into a friend's motivations. Increased uncertainty about others' thoughts could therefore increase directed exploration online, such as exploring for information on social media or ruminating (36,95).

ADDRESSING OPEN QUESTIONS IN SOCIAL MEDIA RESEARCH

The translation of human RL findings from the lab to real life is in its infancy (6,94–96). Nevertheless, in the future, characterizing how RL mechanisms shape social media behavior may

Box 3. Extending the RL Framework

Different RL Systems

The high-level RL process characterized in this review spans many more specific distinctions between different RL systems with different architectures. We have discussed how model-based RL, model-free RL, and value-free processes may differentially underlie instances of goal-directed behavior with different levels of planning depth and habitual behavior. However, further exploring the different computational processes [model-based RL, model-free RL, inverse RL (147), successor-representation RL (80), value-free processes, and more] that underlie different social media behaviors would be a fruitful avenue for future research (113).

Modeling Action Speed

The most prominent RL models in the literature involve selecting between multiple actions (5,81). However, our framework also applies to other aspects of action selection, such as selecting how fast to act, which has been modeled in an RL framework both in experiments (148) and on social media (6,83–85). Thus, the action selection stage should be understood as encompassing both which action to select and the speed at which to perform it. Importantly, although response speed models have been relatively neglected in the RL literature, they may be key for understanding compulsive social media use, including persistent actions such as newsfeed scrolling. Future theoretical and empirical RL research should establish how, if at all, the RL biases that we have discussed affect action speed both in experimental and naturalistic environments.

Multiple Goals

While classical RL models typically maximize reward on a single dimension (5), in naturalistic settings, humans are optimizing multiple, sometimes divergent, utility functions including not just social reward but also money, health, and any other goals. Thus, any model of naturalistic behavior that only accounts for social reward may fail to explain apparently divergent actions that are actually optimizing different goals, such as when apparently suboptimal decisions in a monetary reward task can be explained by the interaction between monetary reward and social preferences (129). Recent RL models have started to integrate this perspective. For example, one review describes how goals moderate the subjective value of rewards, such that more is not always better (130). Another model represents reward in a multidimensional goal space (128). Therefore, future work should elaborate how the pursuit of social rewards on social media operates in the context of maximizing overall utility in naturalistic settings.

RL, reinforcement learning.

inspire new ways to approach policy-relevant questions about social media's impact on users and society (97).

Social Media and Mental Health

Time spent on social media is positively associated with mental wellbeing for some adolescents and negatively associated for others (98). However, the mechanisms that underlie such variation, and consequently how to help those most vulnerable, remain unclear (98,99).

Decades of research in offline[2], experimental contexts have associated poorer mental health with distinct RL computational phenotypes, which are specific reward-learning strategies that can be summarized as a collection of RL parameters (100). Digital phenotyping research has already demonstrated the possibility of predicting mental health symptomatology from passive digital trace data (101–103). However, RL computational phenotyping could go a step further, using theory-driven constructs to explain rather than merely predict how mental health relates to social media use (104).

There is an ongoing controversy regarding the reliability and generalizability of RL phenotyping (105–107), but if robust, RL phenotypes may represent specific vulnerabilities in online environments. For example, adult depression is associated with heightened negativity bias (108–110), which could be perpetuated as social media algorithms disproportionately supply information that reinforces (negative) preexisting beliefs (personalization, Table 1). Additionally, adults with depression have increased effort cost (111), which could reduce the subjective value of high-effort offline social rewards relative to low-effort social media rewards (availability and quantifiability, Table 1). This could partially explain why lower life satisfaction predicts more time spent on social media (112), following Hernstein's matching law that time spent on each activity is allocated as a function of how rewarding it is (26).

RL could also advance our understanding of digital self-regulation. Despite much concern about social media addiction (113), progress understanding digital compulsive behavior is stymied by a lack of consensus about how to identify social media addiction, including whether it represents a discrete category at all (114). Most research has used self-report questionnaires adapted directly from the drug addiction literature and focused on characteristics such as time spent online, which is ambiguous as to psychological processes (114). In contrast, in the future, computational modeling could distinguish whether time spent online arises from dysfunctional reward pursuing (e.g., compulsive use) (115) or another process (such as a fulfilling leisure activity) (116). RL models of response speed could provide a particularly useful framework to study persistent social media use (6,84,85). Identifying which RL process underlies social media behavior, for example by inferring RL parameters directly from objective social media data (6,84,85), could bridge social media and drug addiction scenarios to identify compulsive behavior via shared latent cognitive mechanisms.

Social Media Use Across Development

Both social media use and its effects change across development (117); during adolescence, social media use is both particularly high (118) and predictive of poorer subsequent mental health (112). In parallel, offline computational research has demonstrated how the RL phenotype changes with age (56). Therefore, RL phenotypes across development may

[2]Here, we use "offline" to refer to research that is not about or on social media. However, we acknowledge that this research includes both experiments in the laboratory and experiments conducted with participants recruited via online platforms such as Prolific or MTurk.

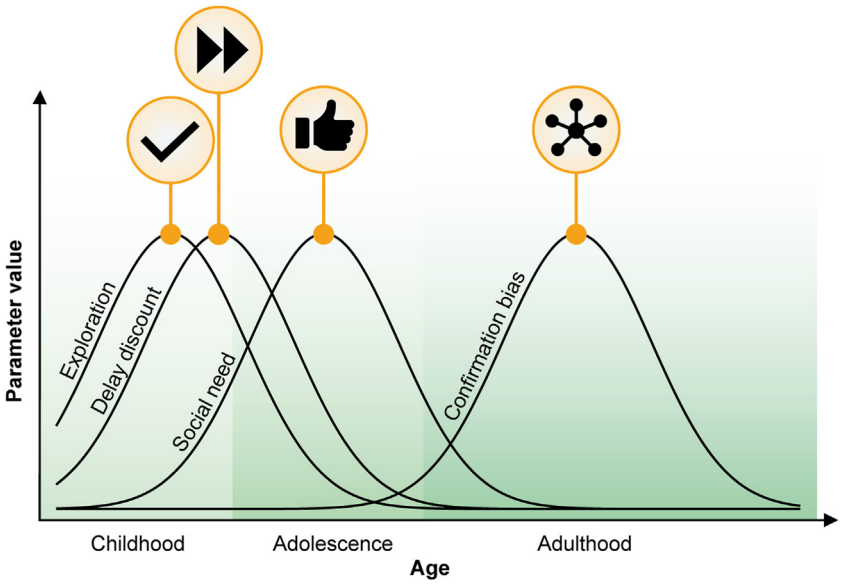

Figure 2. Graph showing interaction between parameter values across development and the vulnerability to specific affordances of social media, as predicted by our framework. The figure represents the approximate peak of each tendency across the life course based on previous research. Future research is needed to precisely delineate the shape and location of the developmental trajectories of each parameter. During childhood, both impulsive directed exploration and delay discounting are higher, which could make young people more vulnerable to affordances that capitalize on these parameters. This includes responding to increased social uncertainty (exacerbated by features such as read receipts, which can make social information more uncertain) by spending large amounts of time consuming reel-type social media structures and fast-forward options for video content. During adolescence, the need for social connection is very high, meaning that proximal rewards on social media, such as likes, may be particularly rewarding. Finally, confirmation bias is higher in adulthood than adolescence, which could make older people particularly vulnerable to forming inaccurate beliefs in response to algorithmic biases that preferentially provide information that confirms users' beliefs.

underlie differential vulnerability to negative and positive effects of different digital affordances (Figure 2).

For example, adolescents have more impulsive, uncertainty-driven directed exploration (56,119) and higher delay discounting (120,121) than adults. Increased uncertainty online could motivate directed exploration for new social information (36). Meanwhile, increased speed of available new social information may interact with high delay discounting to make online rumination more alluring than exploring offline. Together, these could contribute to young people's attraction to social media platforms like TikTok (1), which provide information via fast-forwarded short-form videos.

However, adolescents may also particularly benefit from social media that foster meaningful social connection, given their heightened developmental need for social interaction (a higher set point within social homeostasis) (122–124). Increasing the associative certainty between online proximal rewards and ultimate goals may help maximize this benefit. For example, social media platform design could replace quantifiable online rewards such as likes with clearer indicators of feeling and effort (Table 1).

In contrast, adults have higher confirmation bias than younger people (125) and so may be particularly susceptible to potentially polarizing effects of algorithmic content personalization (60). Indeed, adults are more likely than younger people to believe in and share fake news (126).

Overall, connecting developmental RL and social media research could help illuminate determinants of developmental sensitivity to social media (15,19,112,127). However, we note that our suggestions are mostly based on experimental

Box 4. Outstanding Questions and Future Directions

- Do individual differences in RL parameters mediate individual differences in the drivers and effects of social media use?
- Can knowledge of human RL contribute to wellbeing-centered social media design?
- How can this RL framework be used to understand effects of social media other than wellbeing, for example opinion formation and politics (134)?
- How can RL be used to understand social media interactions at the population as well as the individual level, such as information transmission across social groups or voting patterns?
- How does RL interact with other forms of social learning (such as imitation or advice giving) to shape online behavior?
- Do developmental changes in RL phenotype partially underlie developmental sensitivity to social media use?
- How does RL on social media differ across cultural contexts, particularly in the Global South, which has been relatively understudied in social media research?
- How does social reward maximization on social media relate to behaviors that satisfy other goals, such as other forms of utility?
- How does risk-sensitive RL respond to altered levels of risk associated with social actions on social media?
- How do additional RL biases, such as increased value of options that confer agency or free choice (149), interact with social media features and affordances?
- What are the neurobiological and physiological substrates of RL on social media?

RL, reinforcement learning.

findings that have yet to be extended to naturalistic contexts. Laboratory models of RL have several key simplifications: For example, experiments typically model rewards using numerical points on a single dimension, which neglects the complexity of social rewards and interaction with nonsocial goals in the real world. Using real-world data (6,7,83–85), as well as exploring how multiple goals interact (128–130), may help to close this gap (Box 3). Additionally, most existing research focuses on the Global North (131). However, cultural variation both in responses to offline rewards such as smiles (132) and in the role of social media in daily life (133) means that social media mechanisms likely vary across cultures. Therefore, it will be crucial to consider cultural context in any practical implementation of this research (Box 4).

CONCLUSIONS

Viewing social media behavior as an RL process, humans can be understood as social reward learners adapted to the computational structure of the environments that we developed in (134). Crucially, this helps explain why certain ways of structuring the social (media) environment can better enable the human reward learning system to satisfy social needs. In this review, we have delineated the promise of an RL approach, but we have only scratched the surface of its potential, leaving many routes open for further research that we discuss in Boxes 3 and 4. Ultimately, RL brings together descriptive accounts of human behavior into a mathematically defined, normative process of reward maximization, providing a formal underpinning for investigation into the promises and dangers of social media use (134).

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